

Problem

Determine the current that flows through an $20\text{-}\Omega$ resistor connected to a voltage source $v_s = 120 \cos(377t + 37^\circ)$ V.

Step-by-step solution

Step 1 of 2

Write the sinusoidal expression of voltage source.

$$v_s = 120 \cos(377t + 37^\circ) \text{ V}$$

Here,

The magnitude of the voltage source v_s is 120 V

The angular frequency ω is 377 rad/s

The phase angle is 37° .

The phasor representation of voltage source expression is,

$$\mathbf{V}_s = 120 \angle 37^\circ \text{ V}$$

Step 2 of 2

Calculate the current through $20\text{-}\Omega$ resistor using ohms law.

$$\mathbf{V}_s = \mathbf{I}R$$

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{R} \\ &= \frac{120 \angle 37^\circ}{20} \\ &= 6 \angle 37^\circ \text{ A} \end{aligned}$$

Convert the current expression from phasor domain to time domain.

$$\begin{aligned} i &= i_m \cos(\omega t + \phi) \\ &= 6 \cos(377t + 37^\circ) \text{ A} \end{aligned}$$

Therefore, the current through $20\text{-}\Omega$ resistor is $6 \cos(377t + 37^\circ) \text{ A}$.